Weekly Meeting Update

Sprint Progress & Foundation Milestone

Over the past week, we completed and merged all foundational scaffold tasks across our repositories. Our root builds, container packaging, and local backing services are now operational, allowing us to transition directly into building the Phase 1 functional pipeline.

We also completed code reviews across all active pull requests to ensure cross-repository architectural alignment. Specifically, we standardized our input mount path to `/data` (avoiding collisions with our shared Java module), established our three-tier workload exit code convention (`0`, `10`, `20`), and aligned database primary keys to use 36-character UUID strings matching our queue messages and Kubernetes pod labels.

PR Reviews & In-Flight Work

- **PR #2: Workload Project Scaffold (`batch-workload`) Approved & Merged**
    - Aaron updated the worker to mount input files at `/data`, migrated the build to standard Maven with a Java 17 release target, and implemented a multi-stage Dockerfile. Verified local packaging and flat-memory streaming via `Files.copy`.
- **PR #4: Multi-Module Java 17 Project Scaffold (`job-orchestrator`) Approved & Merged**
    - Set up root parent POM, centralized dependency management (JUnit 5 BOM, SLF4J), Maven wrapper (`mvnw`), and five wired submodules (`common`, `producer`, `consumer`, `status-tracker`, `api`).
- **PR #5: CORE-2 Common Contracts & Schemas (`job-orchestrator`) In Review**
    - Kenny branched off `main` and defined the `JobMessage` record, `ExitCode` enum, and core interfaces (`JobQueue`, `JobRepository`, `ClusterClient`, `StorageClient`). Feedback provided to confirm exit codes and reconcile minor field additions (`schemaVersion`, `fileName`, and `StorageClient.write()`).
- **PR #3: INFRA-2 MySQL V1 DDL Migrations (`platform-infra`) Changes Requested**
    - Andrew drafted the initial relational tables. Review feedback requested updating `job_id` from `BIGINT AUTO_INCREMENT` to `VARCHAR(36)` UUID to match queue contracts, adopting the Flyway double-underscore naming convention (`V1__initial_schema.sql`), and adding a composite primary key on `file_progress`.

New Tasks Added to the Board (`MVP Semester 1`)

With the foundation complete, I added the next five Phase 1 implementation tickets to the project board. These cover the end-to-end data flow from file generation to database insertion:

- **CONSUMER-1: Redis Stream FIFO Dequeue & Concurrency Limiter (`job-orchestrator`)**
  - Implements Redis Streams queue consumer using Lettuce. Dequeues messages in strict FIFO order using consumer groups, validates payload integrity, transitions MySQL state to `DISPATCHED`, and enforces a configurable active pod concurrency limit.
- **SCHED-1: Kubernetes Job Spec Builder & ClusterClient Implementation (`job-orchestrator`)**
  - Implements the `ClusterClient` interface using Fabric8 / Kubernetes Java Client to translate `JobMessage` objects into Kubernetes batch Job specifications. Injects `JOB_ID` and `FILE_PATH`, pins `backoffLimit: 0`, and applies tracking labels.
- **WORK-2: Streamed CSV Parser & Batched MySQL Loading (`batch-workload`)**
  - Upgrades the batch worker to stream CSV files line-by-line without buffering full files in memory. Executes batched JDBC inserts into MySQL `account_payment` (1,000-5,000 rows/batch), checkpoints progress into `file_progress`, and returns appropriate exit codes.
- **PROD-1: Scheduled CSV Data Generator & Redis Stream Publisher (`job-orchestrator`)**
  - Implements a scheduled file generator producing synthetic CSV files matching the Section 6.1 schema. Saves files to storage (`/data` or RustFS/MinIO), logs initial `QUEUED` records in MySQL, and publishes messages to Redis Streams.
- **TRACKER-1: Kubernetes Pod Event Watcher & Status Persistence (`job-orchestrator`)**
  - Builds the status-tracker service to watch Kubernetes API pod events filtered by `job_id` and `attempt` labels. Records lifecycle transitions into `pod_event`, updates the `job` table (`RUNNING`, `SUCCEEDED`, `RETRY_WAIT`), and reconciles state on startup.